

Auditing Annotation Support and Evaluation Sensitivity in Aerial Vehicle Detection

Xinling Wen, Jiahao Zhang, Wenqi Liu, Yangxu Ren, and Yu Chen

Abstract—Aerial detection results are often compared as if their evaluation support were interchangeable. We introduce a provenance-aware audit that reports annotation-support shift, evaluation-policy sensitivity, and candidate-pool rank robustness. The audit covers VisDrone validation, a local UAVDT conversion, AI-TOD validation, and DOTA-v2 validation for structural analysis, totaling 141,347 vehicle boxes. Seven frozen prediction artifacts are replayed over 9,408 prespecified greedy-rule records. Raw coverage is established before class filtering, so an image with no vehicle prediction remains an evaluated empty prediction. A fixed 24-pixel rule retains 73.3%, 95.1%, and 9.7% of vehicle boxes in VisDrone, local UAVDT, and AI-TOD, respectively. AP_{50} bands reach .097 and F_1 bands reach .330. Across the two multi-candidate pools, AP_{50} winners remain unchanged in 12 headline source–rule–policy checks (10 distinct policy outcomes), and greedy and threshold-first Hungarian matching agree on winners in 24 checks (20 distinct outcomes). In paired 1,000-replicate comparisons, every VisDrone AP_{50}/F_1 95% interval favors SAH, whereas every local-UAVDT sequence-proxy interval contains zero. The audit therefore shows how support and evaluator choices can move reported metrics while rank conclusions require common coverage and uncertainty context.

Index Terms—Aerial detection, annotation support, evaluation sensitivity, benchmark auditing, vehicle detection.

I. INTRODUCTION

Aerial detection scores depend on both prediction artifacts and the annotation support against which they are computed. Platform, resolution, ontology, ignored regions, small-object handling, and matching conventions can all change the target of a reported metric. Existing aerial benchmarks are intentionally heterogeneous [1], [2], [3], [4], [5], [6], and label-space differences and dataset bias further complicate comparison [7], [8].

Recent GRSL work addresses efficient high-resolution modeling and validation bias [9], [10]. Error taxonomies and label-quality studies provide complementary evidence [11], [12], [13], [14]. A practical gap remains: cross-artifact detection studies rarely report the support being scored, the sensitivity of a metric to evaluation policy, and the stability of a within-pool ranking under the same declared rules. Small-object handling

This work was supported by Henan provincial research, innovation, education-reform, talent, and collaborative-innovation programs (Grant Nos. 241111213000, 26IRTSTHN036, YJS2024JD48, 2025SJGLX086Y, and 254000510003).

All authors are with the School of Electronics and Information, Zhengzhou University of Aeronautics, Zhengzhou 450046, China. Xinling Wen and Yu Chen are also with the Henan Key Laboratory of General Aviation Technology and the Collaborative Innovation Center of Aeronautics and Astronautics Electronic Information Technology, Henan Province, Zhengzhou 450046, China.

Corresponding authors: Jiahao Zhang (marchematics@gmail.com) and Yu Chen (chatu@zua.edu.cn).

TABLE I

DECLARED ARTIFACT SUPPORT AND RAW-PREDICTION COVERAGE. A COVERED IMAGE WITH NO RETAINED VEHICLE-CLASS PREDICTION IS EVALUATED AS EMPTY.

Source	Eval. images	Vehicle+ images	Covered images	Boxes	Median px side	Median norm. side
VisDrone	548	519	548	17,040	41.0	.0412
Local UAVDT	305	305	305	11,054	33.0	.0278
AI-TOD	2,804	1,443	1,869	59,904	14.0	.0175
DOTA-v2	5,108	2,581	–	53,349	18.0	.0352

makes this gap especially important when pixel support is limited [6].

We address this gap with a three-part audit: (i) *annotation-support shift*, the change in retained valid boxes under a declared support rule; (ii) an *evaluation-policy band*, $\max_r m(r) - \min_r m(r)$ over prespecified rules r ; and (iii) *candidate-pool rank robustness*, agreement of the winner within a declared multi-candidate pool. The protocol makes the scored target explicit before cross-artifact metrics are interpreted.

II. AUDIT DESIGN

A. Artifacts and common coverage

We audit VisDrone2019-DET validation, a local COCO-style UAVDT conversion, AI-TOD validation, and patched DOTA-v2 validation [1], [3], [6], [15]. The available AI-TOD crop reconstruction is xView-derived [5]. The vehicle mapping is VisDrone {car, van, truck, bus}; UAVDT {car, truck, bus}; AI-TOD {vehicle}; and DOTA-v2 {small-vehicle, large-vehicle}. We retain VisDrone source ignores and retain all truncation and occlusion attributes. UAVDT uses the local COCO *xywh* conversion. DOTA quadrilaterals are converted to axis-aligned boxes for structural scale analysis; it contributes no prediction-based evaluation. The provenance manifest records the patched DOTA artifact and all conversion details.

AI-TOD has 1,869 covered images with 16,900 vehicle boxes (9.0/image; median side 12 px), compared with 43,004 boxes in 935 noncovered images (46.0/image; median side 16 px). It therefore contributes a coverage-conditioned, single-candidate sensitivity analysis. Multi-candidate rank robustness is assessed on VisDrone and local UAVDT, which have complete source-wise common coverage (548/548 and 305/305).

Coverage is the presence of an image name in the *raw* prediction artifact before vehicle-class filtering. This distinguishes an executed image with zero retained vehicle predictions from a missing cache record. After filtering, the former remains in the evaluation and contributes valid GT as false negatives. All within-source candidate comparisons use the source-wise

common coverage in Table I; the AI-TOD covered/noncovered contrast is reported as selection evidence.

B. Evaluation contract

For a box (w, h) in image (W, H) , absolute support is $s_a = \max(w, h)$ and normalized support is $s_n = \max(w/W, h/H)$. The prespecified grid contains $s_a \in \{16, 20, 24, 28, 32, 40, 48, 64\}$ px, $s_n \in \{.005, .0075, .010, .015, .020, .030\}$, confidence $\in \{.10, .15, .20, .25, .30, .35, .40, .50\}$, and IoU $\in \{.25, .30, .40, .50\}$. The headline thresholds, 24 px and .015, are interior grid values that expose the absolute/normalized support contrast.

The full surface contains 9,408 greedy records. The AP surface evaluates seven candidates under two scale rules and three policies (42 records), while the matching control uses the same scale rules and policies at confidence .25 and IoU .25/.50 (84 records). For pairwise diagnostics, $\mathcal{R}_h^{AP} = \{(a, 24 \text{ px}), (n, .015)\} \times \Pi$ and $\mathcal{R}_h^{F_1} = \{(a, 24 \text{ px}), (n, .015)\} \times \Pi$ each contain six nominal scale-policy settings; the latter fixes $c = .25$ and $\iota = .25$. The matching set is $\mathcal{R}_h^{\text{match}} = \{(a, 24 \text{ px}), (n, .015)\} \times \Pi \times \{.25, .50\}$ at $c = .25$. Exact ties are treated as rank instability.

The primary policy set is $\Pi = \{I, O, N\}$: I =include-all, O =exclude/source-ignore-off, and N =exclude/source-ignore-on. I retains all mapped valid boxes. O removes below-threshold boxes while retaining predictions as eligible false positives. N uses the same removal and suppresses predictions overlapping source-provided ignore regions at .5 IoU. Under O and N , removed valid boxes retain ordinary non-ignore status. An ancillary *exclude-and-ignore-removed* rule suppresses predictions on removed boxes as well as source ignores, isolating the effect of removed-object treatment.

Greedy matching assigns each score-ordered prediction to the best unmatched eligible GT. Threshold-first lexicographic Hungarian matching first maximizes the number of valid IoU-threshold matches and then their total IoU, with dummy nodes for unmatched objects. Fixed-operating-point performance is global micro- $F_1 = 2TP/(2TP + FP + FN)$ over evaluated images. AP uses score-ordered COCO bbox evaluation [16] with common category 1, standard area ranges, and $\text{maxDets}=100$. The one-to-one overlap contract follows standard detection evaluation [17]; cache scores and hashes are retained in the candidate manifest.

For a full rule $r = (d, \tau, \pi, c, \iota)$, the support shift is $1 - |G_s(d, \tau)|/|G_s|$. At fixed (d, τ, c, ι) , the *policy band* is

$$B_m^\pi = \max_{\pi \in \Pi_s} m(d, \tau, \pi, c, \iota) - \min_{\pi \in \Pi_s} m(d, \tau, \pi, c, \iota).$$

Rank robustness is evaluated on multi-candidate pools that share C_s and is the fraction of fixed settings with the same maximizing candidate.

C. Rank diagnostic and uncertainty

For a finite headline set $\mathcal{R}_h$, the candidate-level *rule band* is $B_k(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} m_k(r) - \min_{r \in \mathcal{R}_h} m_k(r)$. For reference rule r_0 and winner k^* , let $\Delta_j(r) = m_{k^*}(r) - m_j(r)$ and

$$\Gamma_j(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} |\Delta_j(r) - \Delta_j(r_0)|.$$

TABLE II
FROZEN CANDIDATE POOL AND INCLUDE-ALL REFERENCE AP₅₀ POINT ESTIMATES AT $\text{maxDets}=100$. THE POOL CONTAINS EVERY RAW CACHE AVAILABLE AT AUDIT FREEZE.

Source	Frozen pool	Reference AP ₅₀ @100	Role
VisDrone	baseline640; SAHI640	SAHI .718 / base .364	two-variant pool
Local UAVDT	baseline; tiling; YOLO11n; YOLO11m	tiling .249 / base .215	four-variant pool
AI-TOD	baseline640	baseline .234	single-candidate sensitivity
DOTA-v2	none	–	structural-only

This differential radius measures policy-induced movement of the pairwise gap. Normalized support is invariant under isotropic resizing. If $\Delta_j(r_0) > \Gamma_j(\mathcal{R}_h)$ for every competitor j , then k^* remains the winner on $\mathcal{R}_h$; moreover, $\Gamma_j(\mathcal{R}_h) \leq B_{k^*}(\mathcal{R}_h) + B_j(\mathcal{R}_h)$. Proofs and complementary bounds are provided in the supplementary material.

We use a paired 1,000-replicate bootstrap (seed 20260712) of the leading and runner-up candidates. Resampling units are images for VisDrone and filename-derived sequence proxies for local UAVDT; the latter yields 93 units over 305 images (min/median/max unit sizes 1/1/104). Each replicate recomputes global F_1 and score-ordered one-category COCO AP₅₀ (IoU .5, $\text{maxDets}=100$), preserving candidate pairing. Percentile 95% intervals, point differences, and winner probabilities are released in the supplementary repository. The UAVDT result is interpreted as a sequence-proxy uncertainty analysis.

III. RESULTS

Figure 1 links the evaluation denominator to the target-set shift. At 24 px, VisDrone, local UAVDT, and AI-TOD retain 73.3%, 95.1%, and 9.7% of boxes; at .015 they retain 89.8%, 97.4%, and 81.0%. The absolute-rule curves show that one pixel cutoff selects markedly different targets across sources. Supplementary Fig. S1 reports the corresponding normalized-support Wasserstein distances.

The 24-px excluded fractions are 26.7% [24.4, 29.2], 4.9% [0.4, 7.0], and 90.3% [89.3, 91.3] for VisDrone, local UAVDT, and AI-TOD (cluster-bootstrap 95% intervals). At confidence/IoU .25, the associated primary F_1 policy bands are .093, .015, and .330; normalized-rule bands are .061, .130, and .136. For AI-TOD, the ancillary exclude-and-ignore-removed rule gives $F_1 = .130$ at 24 px and .317 at .015, compared with .065 and .260 under exclusion without protection. Its policy band therefore reflects both the scale of support removal and the treatment of predictions on removed targets.

Figure 2 places the headline results in their full sensitivity surfaces. AI-TOD shows broad high-sensitivity regions under absolute filtering, local UAVDT becomes strongly sensitive at larger normalized thresholds, and VisDrone changes more gradually. Maximum absolute/normalized bands are .429/.235 for VisDrone, .512/.493 for local UAVDT, and .464/.368 for AI-TOD. The surfaces describe each frozen candidate within its declared source artifact.

The AP headline comprises 42 candidate records at $\text{maxDets}=100$; Table II gives the include-all AP₅₀ reference values. SAHI wins every VisDrone setting and tiling wins every local-UAVDT setting across the 12 nominal source-rule-policy checks. Local UAVDT's two source-ignore policies coincide,

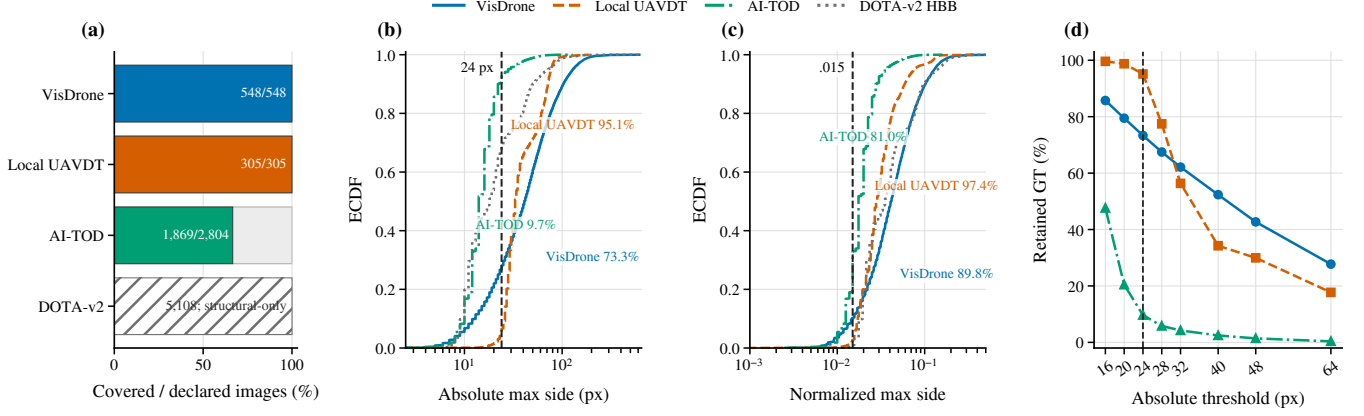


Fig. 1. Dataset support and coverage structure. (a) Source-wise raw-prediction coverage; DOTA-v2 contributes structural-scale evidence. (b),(c) Equal-box-weight ECDFs of absolute and normalized maximum side; labels give support retained at the 24-px and .015 headline thresholds. DOTA-v2 uses an axis-aligned HBB proxy derived from quadrilaterals. (d) Retained-GT curves over the complete absolute threshold grid for the three evaluated sources.

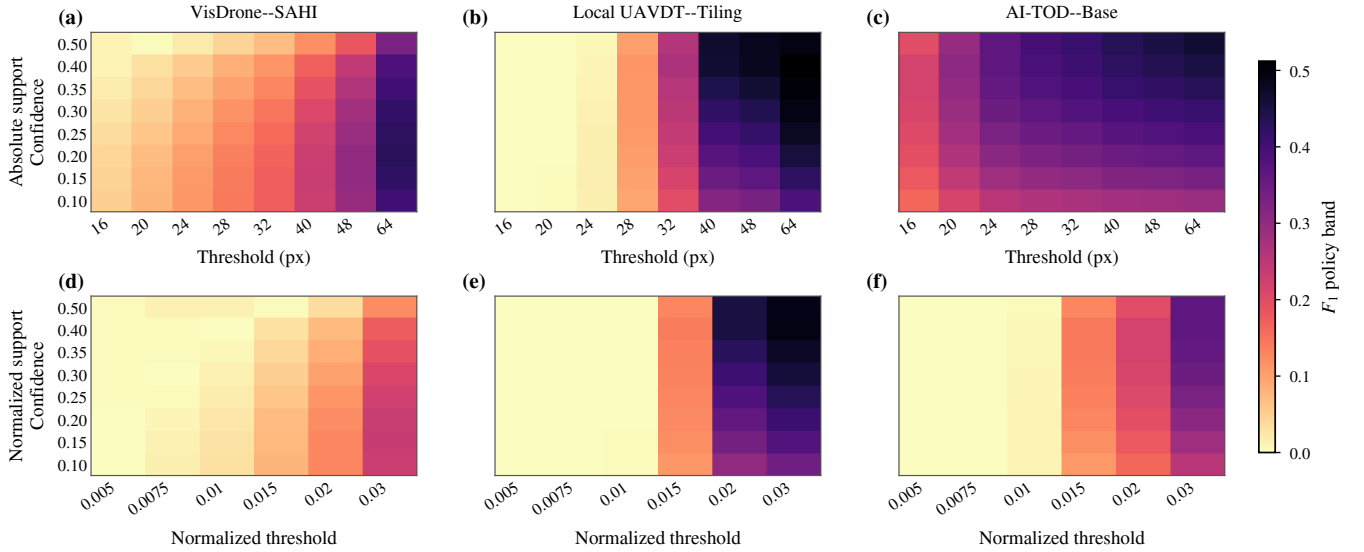


Fig. 2. Full F_1 evaluation-sensitivity surfaces at IoU .25. Columns use the declared VisDrone--SAHI, local-UAVDT--tiling, and coverage-conditioned AI-TOD--baseline artifacts; rows use absolute and normalized support. Each cell is $\max_{\pi \in \Pi} F_1 - \min_{\pi \in \Pi} F_1$ over the three primary policies at a fixed support threshold and confidence. All panels share one color scale.

yielding 10 distinct outcomes. AP_{50} bands span .001–.097 in local UAVDT and .023–.046 in VisDrone.

The paired bootstrap adds uncertainty context to this deterministic result. Across six VisDrone settings, AP_{50} differences have 95% intervals from .339–.362 to .366–.387, all favoring SAHI. Across four local-UAVDT settings, every sequence-proxy interval contains zero (lower endpoints $-.310$ – $-.296$, upper endpoints .071–.111; winner probability .471–.604). Detection-cap sensitivity is also material: after source-ignore handling, $\max Dets=100$ truncates predictions in 406–490 of 548 VisDrone images and 272–292 of 305 local-UAVDT images. At $\max Dets=2000$, no retained prediction is truncated (largest per-image count 1,066), the pairwise winner is unchanged in all 10 distinct outcomes, and AP_{50} gaps shift by $[-.041, .048]$ relative to 100.

Across 84 greedy/Hungarian F_1 records, all 24 multi-candidate checks have the same winner; 20 remain after duplicate local-UAVDT source-ignore outcomes are collapsed.

The maximum absolute F_1 difference is .00546 (Supplementary Fig. S2). Figure 3 complements this deterministic agreement with paired uncertainty: all VisDrone AP_{50}/F_1 intervals are positive, whereas each local-UAVDT sequence-proxy interval contains zero.

A. Ranking under policy perturbation

Figure 3(a) separates metric sensitivity from pairwise-rank movement. All four points satisfy the deterministic certificate on their declared headline rule sets. VisDrone AP_{50} has a SAHI-baseline reference margin of .354 and a differential radius of .021; its observed minimum margin remains .354 because the gap increases under the relevant rules. Local UAVDT has an AP_{50} reference margin of .034 and $\Gamma = .033$, leaving a smallest tiling-baseline margin of .0014. Its deterministic certificate holds, while the sequence-proxy intervals leave the local advantage unresolved. For F_1 , local UAVDT has

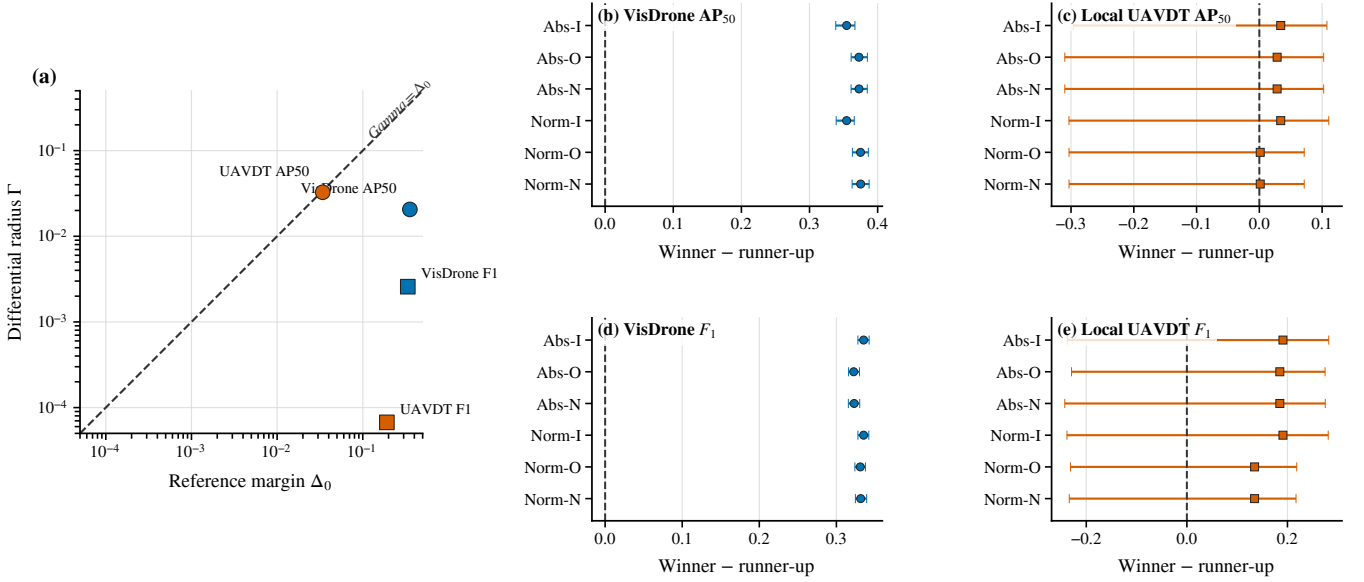


Fig. 3. Rank robustness and paired uncertainty. (a) Deterministic geometry on logarithmic axes: all declared pairwise points satisfy $\Gamma < \Delta_0$. (b)–(e) Paired-bootstrap winner-runner-up differences with 95% intervals for six nominal headline scale-policy settings at AP₅₀ or F₁ (confidence/IoU .25 for F₁). I/O/N denote include-all, exclude/source-ignore-off, and exclude/source-ignore-on. Local-UAVDT AP₅₀ N rows duplicate O because source ignores have no effect in that artifact.

a differential radius of .00007 and a minimum margin of .191. Rank robustness is governed by movement of the pairwise margin; individual policy bands alone are insufficient.

This distinction follows from common-mode response. A rule can lower two candidates by the same amount, producing large individual bands while leaving $\Delta_j(r)$ unchanged. Conversely, anti-aligned small shifts can reverse a close ranking. The conservative bound $B_{k^*}(\mathcal{R}_h) + B_j(\mathcal{R}_h)$ ignores this correlation, whereas $\Gamma_j(\mathcal{R}_h)$ measures its resulting pairwise movement. We therefore use policy bands to describe metric sensitivity, differential radii to diagnose deterministic rank robustness, and the paired bootstrap to quantify uncertainty for the declared resampling unit.

B. Common coverage anchors comparison

Common coverage anchors the ranking comparison. If candidates A and B are observed on different image subsets C_A and C_B , their full-source ordering depends on unobserved outcomes. Shared coverage removes that ambiguity from the declared comparison. Accordingly, AI-TOD contributes structural and policy-sensitivity evidence, while VisDrone and local UAVDT provide the multi-candidate rank analysis. Covered images with no vehicle predictions remain in the latter comparisons. A constructive proof and an F_1 count-perturbation bound are provided in the supplement.

IV. DISCUSSION AND REPRODUCIBILITY

The audit yields a practical reporting template: disclose artifact lineage and image coverage, report a policy band with an explicit removed-object treatment, and pair common-coverage rank results with uncertainty. On shared source-wise image sets, count MAE is 9.0/62.5 for VisDrone SAHI/baseline and 16.8/89.4 for local-UAVDT tiling/baseline.

The frozen pool contains all cached raw-prediction artifacts available at audit freeze: two VisDrone variants (baseline and SAHI [18]), four local UAVDT variants (baseline, tiling, YOLO11n, YOLO11m), and one AI-TOD baseline. This pool defines the reported rank comparisons. Code, frozen manifests, intermediate evaluation records, figure-source data, and SHA-256 checksums are available at <https://anonymous.4open.science/r/aerial-evaluation-audit-5565/README.md>. Together, these records make the evaluation denominator and empty-prediction handling directly inspectable.

V. CONCLUSION

Cross-source aerial evaluation benefits from reporting support shift, policy sensitivity, and rank robustness separately. In these declared artifacts, a 24-px rule retains radically different support across sources, while normalized filtering narrows that disparity. Evaluation rules can move AP₅₀/F₁ substantially even when deterministic winners persist; paired uncertainty can still leave a local advantage unresolved. The practical recommendation is to publish the image-coverage denominator, removed-object treatment, matching contract, and uncertainty-qualified candidate-pool result alongside each cross-source comparison.

REFERENCES

- [1] P. Zhu, L. Wen, X. Bian, H. Ling, and Q. Hu, “Vision meets drones: A challenge,” *arXiv preprint arXiv:1804.07437*, 2018.
- [2] P. Zhu, L. Wen, D. Du, X. Bian, H. Fan, Q. Hu, and H. Ling, “Detection and tracking meet drones challenge,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 11, pp. 7380–7399, 2021.
- [3] D. Du, Y. Qi, H. Yu, Y. Yang, K. Duan, G. Li, W. Zhang, Q. Huang, and Q. Tian, “The unmanned aerial vehicle benchmark: Object detection and tracking,” in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 370–386.

- [4] G.-S. Xia, X. Bai, J. Ding, Z. Zhu, S. Belongie, J. Luo, M. Datcu, M. Pelillo, and L. Zhang, "Dota: A large-scale dataset for object detection in aerial images," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 3974–3983.
- [5] D. Lam, R. Kuzma, K. McGee, S. Dooley, M. Laielli, M. Klaric, Y. Bulatov, and B. McCord, "xview: Objects in context in overhead imagery," *arXiv preprint arXiv:1802.07856*, 2018.
- [6] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized wasserstein distance and a new benchmark," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 190, pp. 79–93, 2022.
- [7] X. Zhao, S. Schuler, G. Sharma, Y.-H. Tsai, M. Chandraker, and Y. Wu, "Object detection with a unified label space from multiple datasets," in *European conference on computer vision*. Springer, 2020, pp. 178–193.
- [8] A. Torralba and A. A. Efros, "Unbiased look at dataset bias," in *CVPR 2011*. IEEE, 2011, pp. 1521–1528.
- [9] E. Zhu, Z. Chen, D. Wang, H. Shi, X. Liu, and L. Wang, "UNetMamba: An efficient UNet-like Mamba for semantic segmentation of high-resolution remote sensing images," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1–5, 2025.
- [10] G. Liu, J. Wen, D. You, Y. Tang, Y. Han, and Q. Li, "Impacts of DEM geolocation bias on multiscale validation of land surface albedo over rugged terrain," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1–5, 2025.
- [11] D. Bolya, S. Foley, J. Hays, and J. Hoffman, "Tide: A general toolbox for identifying object detection errors," in *European Conference on Computer Vision*. Springer, 2020, pp. 558–573.
- [12] C. G. Northcutt, A. Athalye, and J. Mueller, "Pervasive label errors in test sets destabilize machine learning benchmarks," *arXiv preprint arXiv:2103.14749*, 2021.
- [13] J. Murrugarra-Llerena and C. R. Jung, "Noise-aware evaluation of object detectors," in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 2025, pp. 9322–9331.
- [14] M. Schubert, T. Riedlinger, K. Kahl, D. Kröll, S. Schoenen, S. Šegvić, and M. Rottmann, "Identifying label errors in object detection datasets by loss inspection," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2024, pp. 4582–4591.
- [15] J. Ding, N. Xue, G.-S. Xia, X. Bai, W. Yang, M. Y. Yang, S. Belongie, J. Luo, M. Datcu, M. Pelillo *et al.*, "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 11, pp. 7778–7796, 2021.
- [16] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European conference on computer vision*. Springer, 2014, pp. 740–755.
- [17] M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman, "The pascal visual object classes (voc) challenge," *International journal of computer vision*, vol. 88, no. 2, pp. 303–338, 2010.
- [18] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in *2022 IEEE international conference on image processing (ICIP)*. IEEE, 2022, pp. 966–970.